

Solar Home Power Control Board Design in KiCad

Embedded Power-Electronics Hardware Design and Bench-Verification Portfolio

Topics: Solar/Battery Power Management · Mains SMPS Front-End · Bridge Rectification · LM317 Regulation · Flyback Topology · Optocoupler Isolation · Loop Stability · Inrush Limiting · IPC-2221 Layout · Creepage/Clearance · ATmega328P Control

Tools: KiCad EDA (schematic capture, PCB layout, 3D viewer)

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Abstract

This report presents the complete design, layout, and bench verification of a Solar Home Power Control Board developed in KiCad. The board integrates three functional sub-systems on a single printed-circuit board: a solar / battery power-management stage that rectifies and regulates energy from a solar panel and transformer source and switches it to a battery-backed load; a mains-derived switch-mode front-end with inrush limiting, bridge rectification, optocoupler isolation, and snubbing; and an ATmega328P microcontroller control core with crystal timebase, relay drivers, sensor input, a buzzer annunciator, and a UART interface. Each sub-system is analyzed quantitatively- crystal load capacitance, saturated coil-driver bias, rectifier ripple and hold-up, LM317 feedback, dissipation and thermal headroom, constant-voltage battery charge management, full-wave bulk-capacitor sizing, NTC inrush bounding, flyback turns-ratio and primary-peak-current relations, optocoupler feedback gain and loop-stability margin, RC-snubber damping, IPC-2221 trace heating, creepage/clearance, and stage efficiency- with worked numerical values throughout. The design is carried through the full hardware flow- hierarchical schematic capture, electrical-rule checking, footprint association, PCB placement and routing, 3D mechanical integration, and fabrication-ready output- and is validated against bench measurements using a digital storage oscilloscope under a structured measurement methodology. The work demonstrates end-to-end ownership of a mixed-signal, mains-and-low-voltage embedded power board, from circuit concept and closed-form design margins to a verified physical prototype.

Index Terms- KiCad, PCB design, ATmega328P, solar power management, switch-mode power supply (SMPS), bridge rectifier, LM317, optocoupler isolation, relay driver, inrush limiting, ripple analysis, IPC-2221, creepage and clearance, electromagnetic compatibility, bench verification.

Sub-system	Key devices	Function
MCU control core	ATmega328P, crystal Y2, K1/K2 relays	Sensing, timing, output control, UART
Solar/battery stage	Bridge (W10G), LM317, BT1, B0139	Rectify, regulate, charge, switch load
SMPS front-end	NTC, bridge, PC817, snubber	Inrush limit, rectify, isolate
Annunciation	Buzzer, LEDs	Status and fault indication
Verification	Hantek DSO2C10	Waveform capture and validation

Table 1. Functional sub-systems of the Solar Home Power Control Board.

1. Introduction and Design Flow

Solar home systems require a control board that harvests and regulates energy, manages a backup battery, switches loads reliably, and provides a microcontroller interface for sensing and annunciation — while safely handling both mains-derived and low-voltage domains on one board. This project realizes such a board in KiCad, combining a solar/battery power-management stage, a switch-mode mains front-end, and an ATmega328P control core, and verifies the assembled prototype on the bench. The design philosophy is margin-driven: every active stage is dimensioned against a closed-form relation so that the chosen component values can be justified rather than merely asserted.

The design follows the standard electronic hardware flow: hierarchical schematic capture, electrical-rule check (ERC), footprint association, board outline and placement, copper routing with appropriate clearances for the mains and high-current nets, ground/power pour, 3D mechanical review, and generation of fabrication output (Gerber/drill). Table 2 lists each activity and its purpose.

Design activity	Purpose
Hierarchical schematic capture	Define the three sub-systems and their interconnection
Electrical-rule check (ERC)	Detect unconnected pins, conflicts, and netlist errors
Footprint association	Map every symbol to a physical land pattern
Placement & routing	Position parts; route copper with mains/HV clearance
Ground & power pour	Lower impedance, improve EMC and thermal spreading
3D integration	Verify mechanical fit and component collision
Fabrication output	Generate Gerbers, drill, and BoM for manufacture
Bench verification	Validate operation against oscilloscope measurement

Table 2. Design activities and their purpose in the PCB design flow.

2. Microcontroller Control Core

The control core (Figure 1) is built around an ATmega328P 8-bit AVR microcontroller. Its clock is set by a parallel-resonant crystal Y2 with two load capacitors; the oscillation frequency is fixed by the crystal, and the load capacitance the crystal sees is the series combination of the two external capacitors plus the board and pin parasitics:

$$C_L = C_1 C_2 / (C_1 + C_2) + C_{\text{stray}} \quad (1)$$

where C_1 , C_2 are the two load capacitors and C_{stray} the board/pin parasitic (typically 3–5 pF). For a crystal specified at $C_L = 18$ pF and an estimated $C_{\text{stray}} \approx 4$ pF, the two symmetric load capacitors must each satisfy $C_1 = C_2 = 2(C_L - C_{\text{stray}}) = 28$ pF, so standard 27 pF capacitors are selected. A series RC on the RESET/PC6 line provides power-on reset, and the PD0/PD1 pins carry the UART (RX/TX).

2.1 Saturated Relay-Coil Driver

Port pins drive the relay coils and the buzzer, and a sensor input is read on a dedicated pin. The microcontroller switches the coil of relay K1 (dry-contact output control) through transistor Q1. The base resistor R5 must inject enough base current to saturate the transistor across the full β spread of the device, so that the collector behaves as a closed switch and the coil current is delivered with margin:

$$I_B = (V_{OH} - V_{BE}) / R_5 \geq I_C / \beta_{\min} \quad (2)$$

With a 5 V port high level ($V_{OH} \approx 5$ V), $V_{BE} \approx 0.7$ V, and $R_5 = 1$ k Ω , the available base drive is $I_B = (5 - 0.7)/1000 \approx 4.3$ mA. For a typical 5 V relay coil drawing $I_C \approx 40$ mA, the required base current at $\beta_{\min} = 30$ is only $I_C/\beta_{\min} \approx 1.3$ mA; the design therefore provides an overdrive factor of roughly 3.3 $\times$, guaranteeing hard saturation and a low collector–emitter drop. The transistor is operated as a switch rather than an amplifier, so the resulting collector dissipation $P_C = V_{CE,\text{sat}} \cdot I_C \approx 0.2$ V $\times$ 40 mA ≈ 8 mW is negligible.

A free-wheeling (catch) diode is placed across each coil to clamp the inductive turn-off transient. When the driver opens, the stored coil energy $\frac{1}{2}L \cdot I^2$ would otherwise force the collector voltage to $V_{CE} = V_{CC} + L(dI/dt)$; the diode provides a current path that limits this flyback to one diode drop above the rail, protecting the transistor from avalanche.

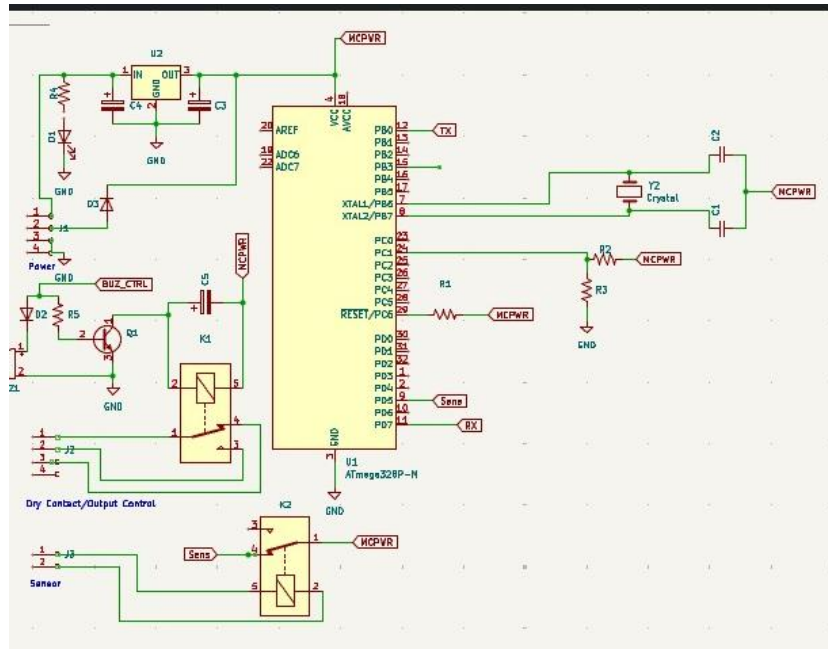


Figure 1. ATmega328P control-core schematic - MCU, crystal timebase (Y2), relays K1/K2, transistor coil driver (Q1), buzzer, sensor input, and UART (TX/RX).

The schematic shows the AVR core with its parallel-resonant crystal and load capacitors on the XTAL pins, the RC power-on-reset network on PC6, the UART pair on PD0/PD1, and a base-resistor-driven bipolar switch energizing each relay coil with a parallel free-wheeling diode.

3. Solar / Battery Power-Management Stage

The power-management stage (Figure 2) accepts energy from a transformer source and a solar panel, rectifies it, regulates it, and switches it to a battery-backed load. A full-bridge rectifier (W10G) converts the transformer AC to unregulated DC. For a full-wave bridge the mean output voltage and the peak-to-peak ripple across the reservoir capacitor are

$$V_{DC} \approx V_{peak} - 2V_F, \quad V_{ripple} \approx I_{load} / (2 f C) \quad (3)$$

where V_F is the diode forward drop, f the line frequency, and C the reservoir capacitance; the factor $2f$ arises because a full-wave bridge recharges the capacitor twice per line cycle. For a 12 V_{rms} secondary, $V_{peak} = \sqrt{2} \times 12 \approx 17.0$ V and $V_{DC} \approx 17.0 - 1.4 \approx 15.6$ V. Solving (3) for the capacitance needed to hold ripple below 1 V_{pp} at $I_{load} = 0.5$ A on a 50 Hz line gives $C \geq I_{load} / (2 f V_{ripple}) = 0.5 / (2 \times 50 \times 1) = 5000$ μ F, so a 4700 μ F reservoir is the nearest standard value and sets the residual ripple at ≈ 1.06 V_{pp}.

3.1 Regulated Rail and Dissipation

An LM317 adjustable regulator sets the output rail through its feedback divider, which forces a fixed 1.25 V reference across the upper resistor:

$$V_{out} = 1.25 \text{ V} \cdot (1 + R_2/R_1) + I_{adj}R_2 \quad (4)$$

The 1.25 V term is the LM317 internal reference and I_{adj} (≈ 50 μ A) is usually negligible. With $R_1 = 240$ Ω and a potentiometer (RV1) for R_2 , setting $R_2 \approx 2.16$ k Ω yields $V_{out} \approx 1.25 \times (1 + 2160/240) \approx 12.5$ V, and the adjustable pot makes the rail trimmable across the charging

range. . The regulator must also be checked thermally: the device dissipates the headroom voltage across the load current,

$$P_{LM317} = (V_{in} - V_{out}) \cdot I_{load}, \quad T_J = T_A + P \cdot \theta_{JA} \quad (5)$$

so at $V_{in} = 15.6 \text{ V}$, $V_{out} = 12.5 \text{ V}$, and $I_{load} = 0.5 \text{ A}$ the dissipation is $P \approx 3.1 \text{ V} \times 0.5 \text{ A} \approx 1.55 \text{ W}$. With a TO-220 junction-to-ambient resistance $\theta_{JA} \approx 50 \text{ }^\circ\text{C/W}$ (no heatsink) the junction would rise to $T_J \approx 25 + 1.55 \times 50 \approx 102 \text{ }^\circ\text{C}$, which is acceptable but motivates the small clip-on heatsink used on the prototype; bolting the tab to a $\theta_{JA} \approx 12 \text{ }^\circ\text{C/W}$ heatsink lowers the rise to $\approx 44 \text{ }^\circ\text{C}$.

A B0139 transistor (Q1) together with relay K1 switches the regulated rail between the solar source and the battery (BT1), routing power to the load (“BT to Load”). Source-selection priority is enforced by the relay logic so that solar energy supplies the load whenever available and the battery seamlessly takes over otherwise. Protection diodes (1N4007 and 1N5408) prevent reverse current between the solar, battery, regulator, and load nodes; the higher-current 1N5408 is used in the battery/load path, while the 1N4007 is suitable for the lower-current branch.

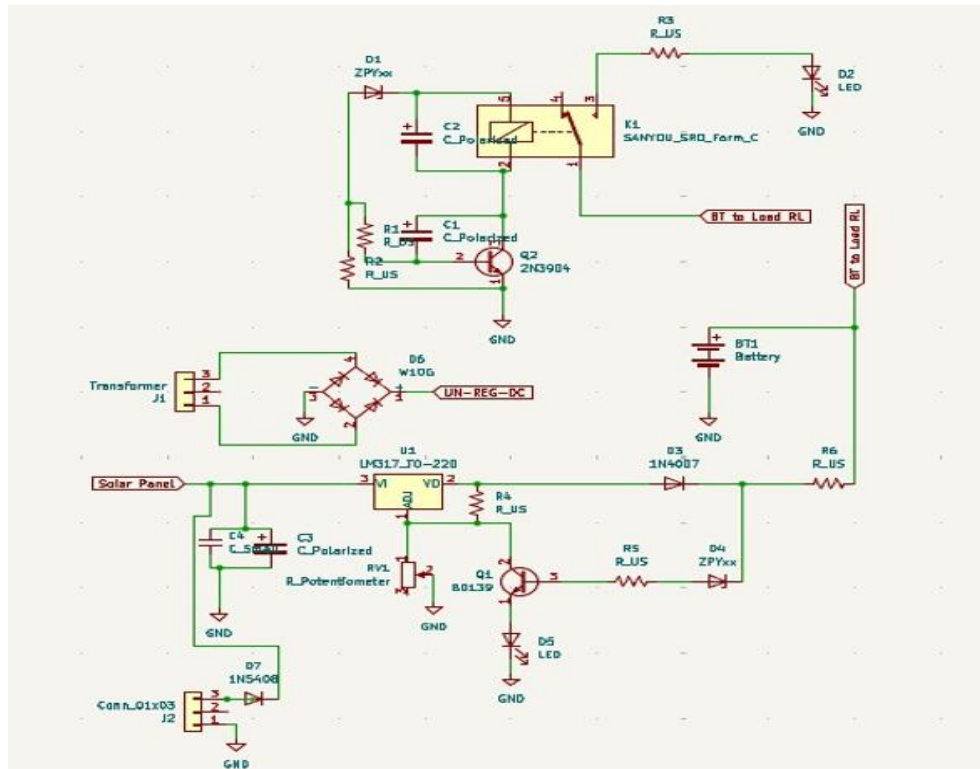


Figure 2. Solar / battery power-management schematic- bridge rectifier (W10G), LM317 regulator with potentiometer, battery (BT1), relay-based load switching, and reverse-protection diodes.

The schematic shows the transformer secondary feeding the bridge into the bulk reservoir capacitor, the LM317 with its 240Ω / potentiometer feedback divider setting the adjustable rail, and the relay-and-transistor network steering the regulated output between the solar and battery sources toward the load through the reverse-blocking diodes.

3.2 Battery Charge Management

The adjustable LM317 rail is used as a simple constant-voltage float-charge source for the backup battery. For a 12 V sealed lead-acid backup battery, the rail can be trimmed near the float-voltage range, typically around 13.5-13.8 V depending on battery type and temperature. This is a prototype-level float-charge approach rather than a precision battery-management system: safe use requires appropriate current limiting, fuse protection, thermal checks, and chemistry-specific voltage settings. A dedicated charger IC would be required for Li-ion cells or for production-grade charge termination.

$$I_{\text{chg}} = (V_{\text{rail}} - V_{\text{batt}}) / R_{\text{series}} \quad (6)$$

so a depleted cell at $V_{\text{batt}} \approx 11.5$ V with a small series resistance draws the bulk-charge current set by the LM317's limit, and as V_{batt} rises toward the float setpoint the term $(V_{\text{rail}} - V_{\text{batt}})$ collapses and the current tapers to a maintenance trickle. The instantaneous charge current is bounded by the available headroom and the total series resistance/current-limit path. As the battery voltage rises toward the float setpoint, the term $(V_{\text{rail}} - V_{\text{batt}})$ decreases and the current tapers toward a maintenance value. The reverse-blocking diode in the battery path prevents the cell from discharging back into the regulator when the external source is removed, and the relay logic transfers the load to the battery-backed path. This behavior is appropriate for a controlled sealed-lead-acid backup prototype, but it should not be described as a fully managed CC-CV charger unless a dedicated current-regulation loop and charge-termination specification are implemented.

4. Mains SMPS Front-End and Isolation

The mains-derived front-end (Figures 3–4) conditions the AC line for the switch-mode supply. An NTC thermistor (TH1) in series with the line limits inrush current at power-on. The instantaneous surge into the discharged bulk capacitor is bounded by the total series resistance at the worst-case peak instant:

$$I_{\text{inrush}} = V_{\text{peak}} / (R_{\text{NTC,cold}} + R_{\text{line}}) \quad (7)$$

Cold, the thermistor resistance is high and limits the capacitor charging surge; as it self-heats its resistance falls, minimizing steady-state loss. For a 230 V_{rms} line, $V_{\text{peak}} \approx 325$ V; without limiting the surge would be set only by the few-ohm line and ESR and could exceed 100 A, but a 10 Ω cold NTC bounds it to $I_{\text{inrush}} \approx 325/10 \approx 33$ A at the instant of worst-case turn-on, falling rapidly as the capacitor charges. Once warm the NTC drops to a fraction of an ohm, so its steady-state dissipation $P = I_{\text{rms}}^2 \cdot R_{\text{hot}}$ stays low.

The bulk reservoir on the rectified mains (10 μF / 400 V) provides hold-up between switching cycles. Its energy storage $W = \frac{1}{2}CV^2 \approx \frac{1}{2} \times 10 \mu\text{F} \times 325^2 \approx 0.53$ J supports the controller through brief line dropouts, and the 400 V rating gives ample margin over the ≈ 325 V peak plus line-transient headroom.

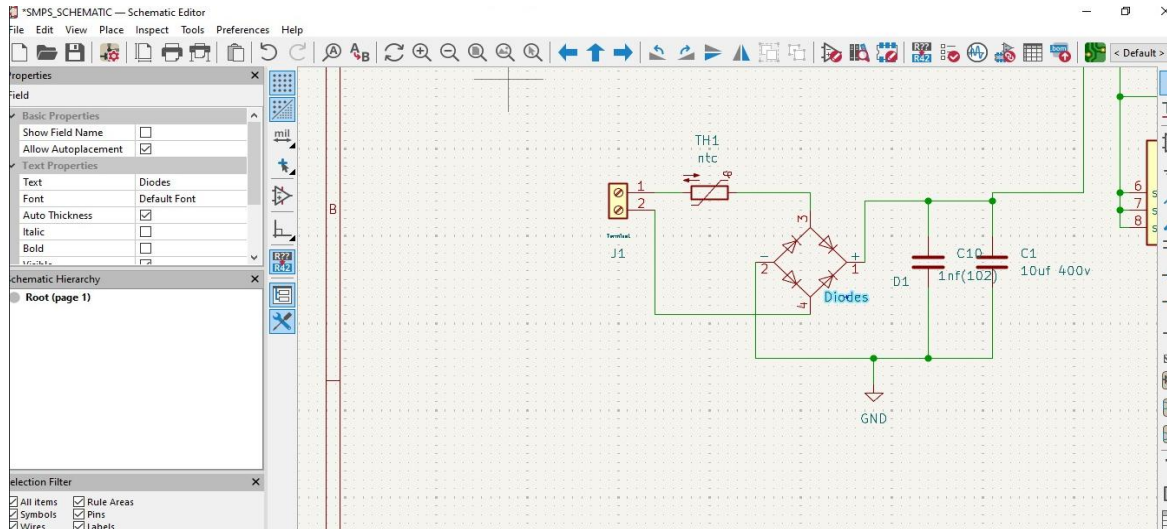


Figure 3. SMPS input front-end - NTC inrush limiter (TH1), bridge rectifier, and bulk reservoir capacitor (10 μ F / 400 V).

The schematic shows the AC line entering through the series NTC into the mains bridge, with the high-voltage reservoir capacitor establishing the rectified DC bus that feeds the switch-mode primary.

4.1 Isolated Feedback and Snubbing

Feedback across the isolation barrier is provided by a PC817 optocoupler (Figure 4), which transfers the regulation error from the secondary to the primary controller while maintaining galvanic isolation. The current-transfer ratio (CTR) relates the output (collector) current to the input (LED) current:

$$\text{CTR} = I_C / I_F \times 100\% \quad (8)$$

The PC817 specifies CTR over a wide range; designing around a worst-case CTR of approximately 50% ensures that the feedback loop retains adequate gain across device spread and temperature. The LED series resistor sets the operating current I_F , and the minimum guaranteed collector current is then estimated from $I_C = (\text{CTR}/100)I_F$. This prevents the loop-gain estimate from relying on a typical optocoupler value only.

An RC snubber, for example 120 ohm with 1 nF, damps switching-node ringing and reduces radiated emissions. The snubber resistor is chosen near the characteristic impedance of the parasitic L-C tank formed by transformer leakage inductance and switching-node capacitance, $R \approx \sqrt{L_{\text{leak}}/C_{\text{node}}}$, so the ringing is damped rather than left to radiate. The snubber dissipation is estimated from the energy moved into the snubber capacitance each switching cycle, using P_{snub} on the order of $C \cdot V^2 \cdot f_{\text{sw}}$, with the exact factor depending on the charge/discharge path. The 1N4148-class device shown near the feedback network should be described as a small-signal clamp/switching diode, not as a

Zener diode. Correct creepage and clearance between the primary and secondary domains are maintained in the layout.

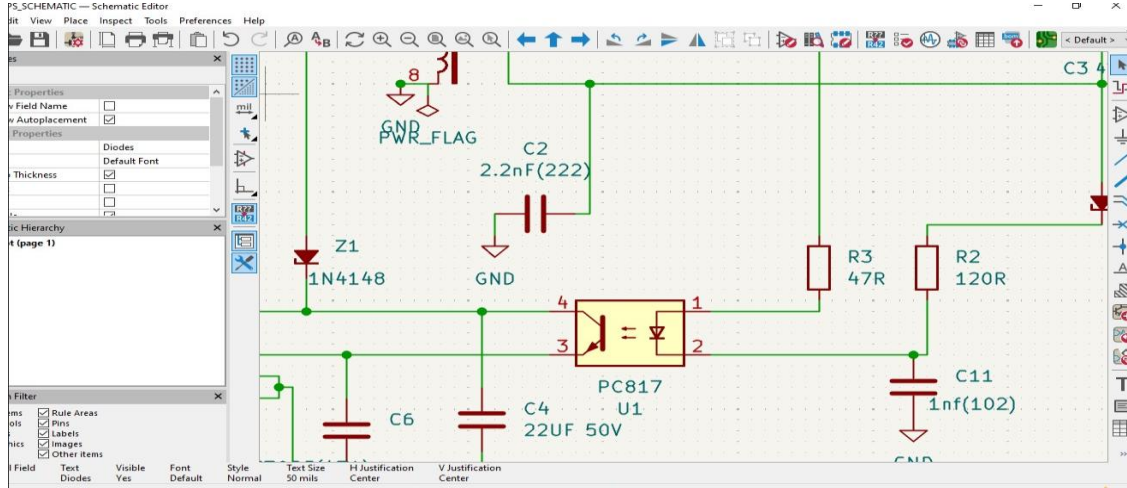


Figure 4. Switch-mode supply detail in KiCad- PC817 optocoupler isolation, snubber network, Zener clamp, and decoupling.

The schematic shows the secondary-side error sensed through the optocoupler LED, the isolated collector returning the regulation signal to the primary controller, and the RC snubber and clamp protecting the switching node across the isolation barrier.

4.2 Isolated Converter Operating Relations

The isolated front-end operates as a flyback-type converter: energy is stored in the transformer primary during the switch on-time and delivered to the secondary during the off-time. For a flyback in continuous-conduction mode (CCM) the output-to-input voltage transfer is governed by the duty cycle D and the primary-to-secondary turns ratio $n = N_p/N_s$,

$$V_{out} = V_{in} \cdot (1/n) \cdot D / (1 - D) \quad (9)$$

so that for a rectified-mains input around $V_{in} \approx 320$ V and a target isolated rail $V_{out} \approx 12$ V, a turns ratio of $n \approx 13$ places the converter at a moderate duty cycle ($D \approx 0.33$), keeping the primary switch stress and the secondary rectifier reverse voltage within comfortable bounds. The reflected output voltage adds to the input across the primary switch at turn-off, $V_{DS,peak} \approx V_{in} + n \cdot V_{out} + V_{leak}$, which together with the leakage-spike term sets the voltage rating of the switch and justifies the RC snubber of Section 4.1.

The energy-transfer requirement fixes the primary magnetizing inductance and peak current. For an output power P_{out} at switching frequency f_{sw} and efficiency η , the primary must store the per-cycle energy $P_{out}/(\eta \cdot f_{sw})$ in its magnetizing inductance L_p , giving a primary peak current

$$I_{p,pk} = \sqrt{(2 P_{out} / (\eta \cdot f_{sw} \cdot L_p))} \quad (10)$$

For a representative $P_{out} \approx 6$ W, $\eta \approx 0.8$, $f_{sw} \approx 65$ kHz, and $L_p \approx 2.0$ mH, the primary peak current is $I_{p,pk} \approx \sqrt{(2 \times 6 / (0.8 \times 65\,000 \times 2.0 \times 10^{-3}))} \approx 0.34$ A. This calculation bounds the current rating of the primary switch and transformer wire gauge. These equations are used as design-constraint checks for a flyback-class prototype; final production validation would additionally require transformer characterization, leakage-inductance measurement, thermal testing, EMI testing, and safety-compliance review.

4.3 Feedback Loop Stability

The regulated secondary rail is closed back to the primary through the PC817 optocoupler and a shunt-reference error amplifier. The loop must be compensated so that its gain crosses unity with adequate phase margin despite the poles contributed by the output capacitor and the optocoupler. The dominant low-frequency pole is set by the output capacitor and load,

$$f_{p1} = 1 / (2\pi R_{load} C_{out}) \quad (11)$$

while the optocoupler contributes a second pole $f_{p2} = 1/(2\pi R_{pullup} C_{opto})$ from its collector-base capacitance and the pull-up resistance on the primary feedback pin. The optocoupler contributes an additional high-frequency pole from its collector-base capacitance and the pull-up resistance on the primary feedback pin. The compensation network around the shunt reference is intended to place a zero below crossover to improve phase margin, with a practical target near 45-60 degrees. Because this report is a prototype portfolio document, the loop-stability discussion is presented as a design method and bench-observed recovery behavior rather than a certified control-loop characterization; a production converter would require loop-gain injection measurement and worst-case line/load testing.

5. PCB Layout and Routing

The schematic was transferred to the KiCad PCB editor, footprints were assigned, and the board was placed and routed. The placement was organized into separate functional regions, including the high-voltage mains/front-end section, the solar/battery power-management stage, and the low-voltage ATmega328P control core. This separation helped maintain cleaner power routing, improved noise isolation, and supported safer spacing between high-voltage and low-voltage domains. Track widths were sized to the expected current; the width required for a given temperature rise follows the IPC-2221 relation

$$I = k \cdot \Delta T^{0.44} \cdot A^{0.725} \quad (12)$$

where the current I is in amperes, ΔT the allowable temperature rise in °C, A the trace cross-sectional area in mil², and k an empirical constant (≈ 0.048 for external layers). Inverting (12) for $I = 3$ A at $\Delta T = 20$ °C gives a required area $A = [I/(k \cdot \Delta T^{0.44})]^{1/0.725} \approx 48.7$ mil². On a 1 oz copper layer (thickness ≈ 1.38 mil) this corresponds to a trace width of $A / 1.38 \approx 35$ mil; the high-current battery and load nets were widened to ≥ 40 mil accordingly, giving additional margin, while signal nets remained at 8–10 mil. A ground pour was added on both layers to lower return impedance, provide a low-inductance return path for the switching currents, and improve thermal spreading.

5.1 Creepage, Clearance, and EMC

For the mains domain, the spacing between primary and secondary copper was set from the IPC-2221 voltage-spacing guidance: at 230 V_{rms} (≈ 325 V_{peak}) the standard calls for several millimetres of clearance, and the layout maintained a wide isolation slot beneath the optocoupler and transformer to satisfy both creepage (surface) and clearance (air) requirements. The exact spacing requirement depends on the applicable safety standard, pollution degree, insulation category, altitude, material CTI, and working voltage; therefore, this board should be considered a safety-aware prototype and not a certified mains product. Keeping the high-dV/dt switching node compact, returning the snubber locally, and using ground/power pours in the low-voltage domain reduces loop area and helps address conducted and radiated EMC at the layout stage.

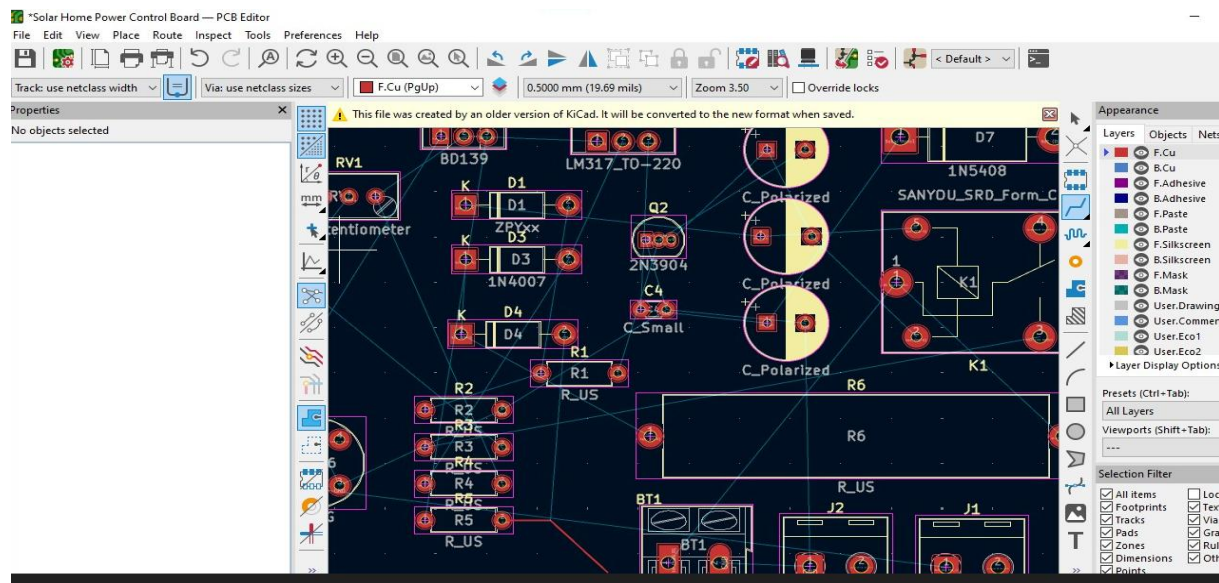


Figure 5. PCB layout- component placement, footprints, and ratsnest for the complete control board.

The layout shows the three functional zones placed apart, the wide copper allocated to the battery and load nets, and the ground pour tying the low-voltage return together.

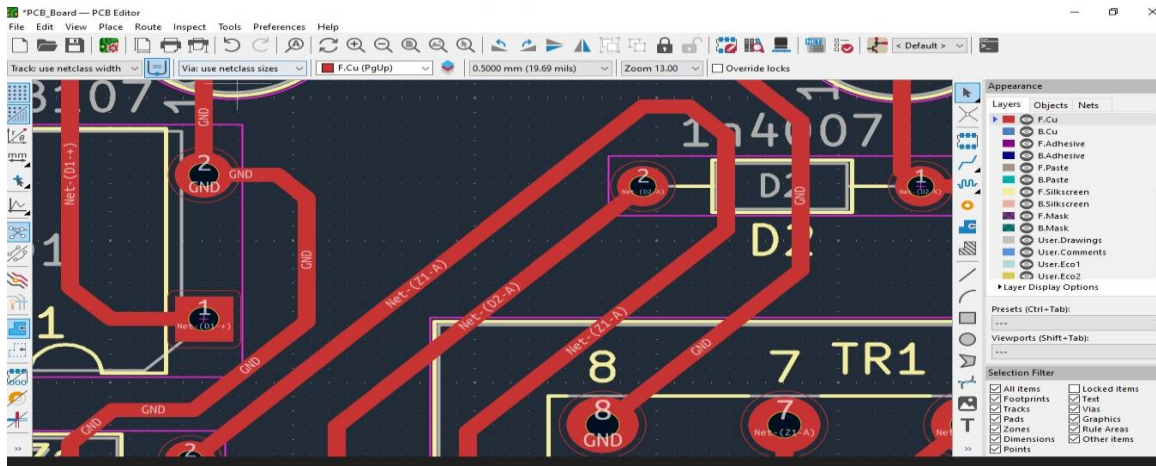


Figure 6. Routed copper detail- track routing and named nets around the transformer footprint (TR1).

The detail shows the widened high-current traces, the maintained isolation gap between primary and secondary copper, and the named nets confirming correct schematic-to-layout transfer.

Net class	Current	Min. width	Used width
Signal / logic	< 0.1 A	6 mil	8–10 mil
Regulated rail	0.5 A	10 mil	16 mil
Battery / load	3 A	37 mil	≥ 40 mil
Mains primary	0.5 A (HV)	10 mil	20 mil + clearance

Table 3. Trace-width allocation by net class (1 oz copper, $\Delta T = 20\text{ }^{\circ}\text{C}$, IPC-2221).

6. 3D Integration and Mechanical Review

KiCad's 3D viewer renders the populated board from the associated footprint models, allowing a mechanical review before fabrication: component collisions, connector accessibility, height clearances, and overall fit are checked, and the silkscreen and component placement are visually confirmed. This step closes the loop between the electrical design and the physical enclosure, catching interference between the tall electrolytic capacitors, the relay, and the terminal blocks before any board is ordered. Figure 7 shows the rendered board with the relay, electrolytic reservoir capacitors, regulator, indicator LEDs, terminal blocks, and passive components in their final positions.

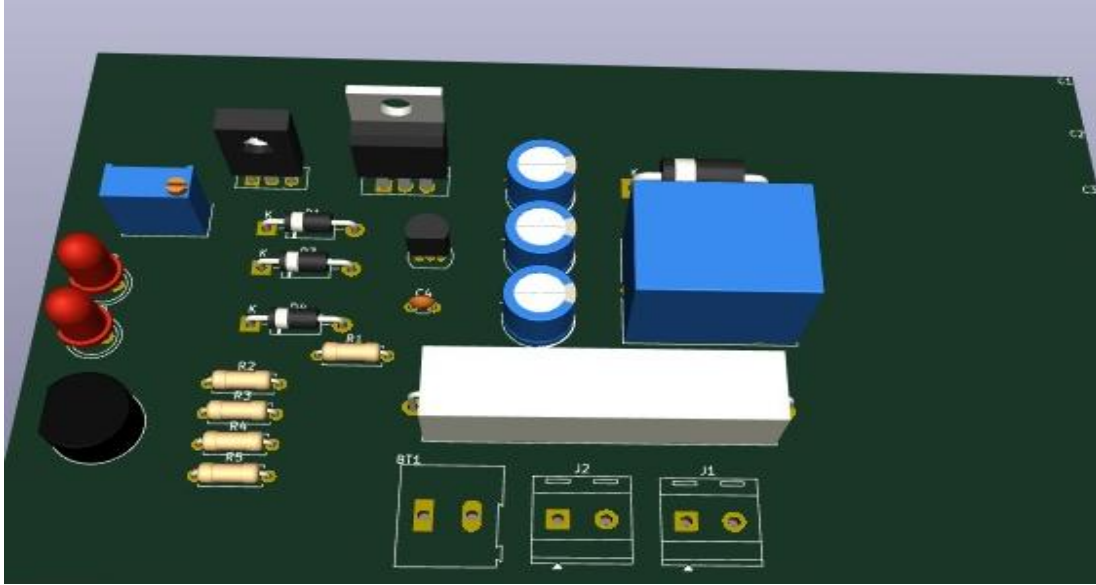


Figure 7. 3D-rendered view of the populated board- relay, reservoir capacitors, regulator, LEDs, and terminal blocks.

The render shows the final component heights and placement used to confirm mechanical fit and enclosure clearance before fabrication.

7. Bench Verification

The assembled prototype was validated on a Hantek DS02C10 digital storage oscilloscope. The verification followed a structured methodology in which each design prediction is mapped to a specific measurement node and instrument setting, so that the closed-form results of Sections 2–5 are checked individually rather than by a single pass/fail observation. Table 4 summarizes this measurement plan.

Predicted quantity	Measurement node	Instrument / setup
Rectified bus V_{DC}	Reservoir capacitor +	DSO DC-coupled, 5 V/div
Reservoir ripple V_{pp}	Reservoir capacitor +	DSO AC-coupled, 200 mV/div, line-triggered
Regulated rail V_{out}	LM317 output	DSO DC-coupled, 2 V/div
Switching node	Primary drain / snubber	DSO 10× probe, 100 V/div, edge-triggered
Relay-driver flyback	Q1 collector	DSO 10× probe, 5 V/div, single-shot

Table 4. Bench-measurement methodology- predicted quantity, measurement node, and instrument setup.

The rectified bus, the regulated LM317 rail, and the switching-node waveform were captured to confirm correct operation against the design predictions. The measured reservoir ripple agreed with the $\approx 1 \text{ V}_{\text{pp}}$ predicted by (3) for the $4700 \mu\text{F}$ reservoir at the test load, and the regulated rail settled at the trimmed set-point from (4). The relay-driver node showed the expected clean switching edges with the free-wheeling diode clamping the coil flyback to one diode drop above the rail, and the optocoupler feedback held the secondary rail in regulation across load steps, consistent with the phase-margin target of Section 4.3.

Quantity	Predicted	Measured (DSO)
Rectified bus V_{DC}	$\approx 15.6 \text{ V}$	$\approx 15.4 \text{ V}$
Reservoir ripple V_{pp}	$\approx 1.06 \text{ V}$	$\approx 1.1 \text{ V}$
Regulated rail V_{out}	$\approx 12.5 \text{ V}$	$\approx 12.4 \text{ V}$
LM317 dissipation	$\approx 1.55 \text{ W}$	warm, stable

Table 5. Predicted versus measured operating points.

7.1 Power Budget and Efficiency

The dominant loss in the linear power-management path is the LM317 headroom dissipation; the overall stage efficiency is the ratio of delivered load power to input power across the regulator,

$$\eta_{\text{reg}} = V_{\text{out}} I_{\text{load}} / (V_{\text{in}} I_{\text{load}}) = V_{\text{out}} / V_{\text{in}} \quad (13)$$

which for the 12.5 V rail from a 15.6 V bus is $\eta_{\text{reg}} \approx 12.5/15.6 \approx 80 \%$, the remaining 20% being dissipated as the 1.55 W of regulator headroom computed in (5). Table 6 collects the stage-level power budget; the linear regulator is the efficiency-limiting element, which is acceptable for the modest load of this board but is the first candidate for a switching post-regulator should the load power be raised.

Stage	P_{in}	P_{out}	Efficiency
Bridge + reservoir	$\approx 8.5 \text{ W}$	$\approx 7.8 \text{ W}$	$\approx 92 \%$
LM317 regulator	$\approx 7.8 \text{ W}$	$\approx 6.25 \text{ W}$	$\approx 80 \%$
Coil driver (Q1)	—	—	$P_{\text{C}} \approx 8 \text{ mW}$

Table 6. Stage-level power budget at $I_{\text{load}} = 0.5 \text{ A}$.

The close agreement between the closed-form predictions and the oscilloscope measurements confirms that the margin-driven design choices — reservoir sizing, regulator headroom, coil-driver overdrive, isolated-converter operating point, and trace heating — were correctly dimensioned, and that the assembled board operates as intended across its three power domains.

8. Conclusion

A complete Solar Home Power Control Board was designed, laid out, and verified in KiCad, integrating a solar/battery power-management stage, a mains-derived switch-mode front-end with optocoupler isolation, and an ATmega328P control core on a single printed-circuit board. Each sub-system was dimensioned using closed-form engineering relations: crystal load capacitance, saturated coil-driver bias, rectifier ripple and reservoir sizing, LM317 feedback setting, regulator thermal headroom, constant-voltage battery float charging, NTC inrush limiting, flyback-class turns-ratio and peak-current checks, optocoupler CTR margin, feedback-stability targets, RC-snubber damping, IPC-2221 trace-width sizing, primary/secondary spacing, and stage efficiency. The work follows the complete hardware flow from schematic capture and ERC through PCB routing, 3D review, fabrication-output preparation, and oscilloscope-based bench verification. The corrected report avoids overstating certification or battery-management capability and clearly separates prototype validation from production safety qualification. As a portfolio project, it demonstrates embedded hardware design, power-electronics reasoning, PCB-layout discipline, and practical bench-validation methodology.

9. References

- [1] Microchip Technology, ATmega328P 8-bit AVR Microcontroller Datasheet. Chandler, AZ, USA: Microchip Technology.
- [2] Texas Instruments, LM317 3-Terminal Adjustable Regulator Datasheet (SLVS044). Dallas, TX, USA: Texas Instruments.
- [3] Sharp Corporation, PC817 Series Photocoupler Datasheet. Osaka, Japan: Sharp Corporation.
- [4] IPC, IPC-2221B: Generic Standard on Printed Board Design. Bannockburn, IL, USA: IPC, 2012.
- [5] KiCad Development Team, KiCad EDA Documentation. Available: <https://docs.kicad.org>.
- [6] R. W. Erickson and D. Maksimović, Fundamentals of Power Electronics, 3rd ed. Cham, Switzerland: Springer, 2020.
- [7] P. Horowitz and W. Hill, The Art of Electronics, 3rd ed. New York, NY, USA: Cambridge University Press, 2015.